

Stellar Interiors, Nucleosynthesis & Evolution

Problem Set — Solutions

fizicyst

Section A — Objective

O1. Answer: (c). $\tau/\tau_{\odot} = (0.5)^{-2.5} = 2^{2.5} \approx 5.66$.

O2. Answer: (b). Iron-56 sits at the summit of the binding-energy-per-nucleon curve, so both fusing it further and splitting it apart require climbing back *up* the curve — neither releases energy.

O3. Answer: (b). $R_s \propto M$, so tripling the mass triples the radius: $3 \times 3 = 9$ km.

O4. Answer: (c) is false. The correct reason neutron stars support more mass is that the neutron's much larger mass pushes the non-relativistic-to-relativistic transition (which sets the effective mass limit) out to far higher densities than for electrons — it isn't about the Fermi momentum itself being smaller. (a), (b), (d) are all correct as stated.

O5. Answer: (d). $2^{18} \approx 2.62 \times 10^5$.

Section B — Subjective

S1.

- (i) Since $R_i \gg R_f$, $U_i \approx 0$, so $|\Delta U| \approx |U_f| = \frac{3GM^2}{5R_f}$. With $GM^2 = 6.674 \times 10^{-11} (1.989 \times 10^{30})^2 \approx 2.64 \times 10^{50}$ and $R_f = 0.01 \text{ pc} \approx 3.09 \times 10^{14} \text{ m}$:

$$|\Delta U| \approx \frac{3(2.64 \times 10^{50})}{5(3.09 \times 10^{14})} \approx 5.1 \times 10^{35} \text{ J}.$$

- (ii) By the virial argument, exactly half goes to heating and half is radiated: $\Delta K \approx 2.6 \times 10^{35} \text{ J}$ retained as heat, and $\approx 2.6 \times 10^{35} \text{ J}$ radiated away.
- (iii) Energy per particle $\approx \Delta K/N \approx (2.6 \times 10^{35})/(1.2 \times 10^{57}) \approx 2.2 \times 10^{-22} \text{ J}$. Setting this to $\frac{3}{2}kT$:

$$T \approx \frac{2(2.2 \times 10^{-22})}{3(1.381 \times 10^{-23})} \approx 10 \text{ K}.$$

This is quite plausible — observed molecular clouds do sit around 10–20 K, so this crude estimate lands in the right ballpark for the *early*, still-diffuse stage of collapse (long before the core gets anywhere near fusion temperatures).

S2.

- (i) Substituting: $P_{\text{deg,rel}} \sim \hbar c \left(\frac{M}{m_H R^3} \right)^{4/3}$. Setting this $\sim GM^2/R^4$:

$$\hbar c \frac{M^{4/3}}{m_H^{4/3} R^4} \sim \frac{GM^2}{R^4}.$$

- (ii) Both sides carry $1/R^4$, which cancels immediately, leaving

$$\hbar c \frac{M^{4/3}}{m_H^{4/3}} \sim GM^2 \implies M^{2/3} \sim \frac{\hbar c}{G m_H^{4/3}} \implies M_{\text{crit}} \sim \left(\frac{\hbar c}{G} \right)^{3/2} \frac{1}{m_H^2}.$$

- (iii) $\hbar c/G \approx (1.055 \times 10^{-34})(3 \times 10^8)/(6.674 \times 10^{-11}) \approx 4.74 \times 10^{-16}$. Then $(\hbar c/G)^{3/2} \approx 1.03 \times 10^{-23}$, and dividing by $m_H^2 \approx 2.80 \times 10^{-54}$:

$$M_{\text{crit}} \sim 3.7 \times 10^{30} \text{ kg} \approx 1.9 M_{\odot}.$$

This is the right order of magnitude and even the right ballpark numerically compared to the actual $1.4 M_{\odot}$ — remarkably good agreement given that every $O(1)$ numerical prefactor (from the detailed relativistic Fermi-gas physics) was simply dropped.

S3.

- (i) $\tau \approx 10 \text{ Gyr} \times 3^{-2.5} \approx 10/15.6 \approx 0.64 \text{ Gyr} \approx 640 \text{ Myr}$.
- (ii) $(1.15)^{17} \approx 10.8$ — roughly an order-of-magnitude jump in energy generation for just a 15% rise in core temperature.
- (iii) Because ε_{CNO} is so ferociously sensitive to T , a star only needs a tiny increase in core temperature to supply a vastly larger luminosity. So as stellar mass climbs and more luminosity is demanded to support a bigger envelope, the core temperature barely has to budge to keep up — the steep T -dependence acts like a thermostat, clamping core temperatures into a narrow band while luminosity (and thus lifetime, via (i)) still swings by orders of magnitude across the mass range.